

Course Name: Introduction to DBMS

Course Code: **Internal Marks: 25** **External Marks: 75** **Credit: 4**

UNIT-I

Introduction:- Characteristics of database approach, data models, DBMS architecture and data independence.

UNIT-II

E-R modeling:- Entity types, Entity set, attribute and key, relationships, relation types, roles and structural constraints, weak entities, enhanced E-R and object modeling, Sub classes; Super classes, inheritance, specialization and generalization.

UNIT-III

File Organization:- Indexed sequential access files; implementation using B & B++ trees, hashing, hashing functions, collision resolution, extendible hashing, dynamic hashing approach implementation and performance.

UNIT-IV

Relational Data Model:- Relational model concepts, relational constraints, relational algebra
SQL: SQL queries, programming using SQL.

UNIT-V

EER and ER to relational mapping:- Data base design using EER to relational language.

UNIT-VI

Data Normalization:- Functional Dependencies, Normal form up to 3 normal form.
Concurrency Control: Transaction processing, locking techniques and associated, database recovery, security and authorization. Recovery Techniques, Database Security.